FORMAL MODULI PROBLEMS AND DIFFERENTIAL GRADED LIE ALGEBRAS

Zhaoqi Zhu

Introduction

Deformation theory studies the behavior of a mathematical object under changes of its parameters near a fixed value. A tangent vector describes a first-order change, but it does not by itself determine whether that change extends to second or higher order. Automorphisms introduce a second difficulty: the collection of deformations is generally a groupoid rather than a set. Derived deformation theory incorporates both phenomena by allowing derived test algebras and by recording spaces of deformations, including their higher automorphisms.

This thesis is an exposition of the first two chapters of Jacob Lurie's *Derived Algebraic Geometry X: Formal Moduli Problems* [Lur11]. Its central objective is the characteristic-zero equivalence

$$\mathrm{Lie}_k \simeq \mathrm{Moduli}_k$$

between differential graded Lie algebras and formal moduli problems over a field k. The correspondence identifies the underlying complex of the Lie algebra controlling X with $T_X[-1]$. The shift is part of the statement: the tangent complex and the controlling Lie algebra have different degree conventions even before the bracket is considered.

The exposition follows Lurie's route through abstract deformation contexts, rather than taking a Maurer–Cartan construction as the definition of the correspondence. Chapter 1 develops the axioms for formal moduli problems and explains the general reconstruction theorem. Chapter 2 constructs the Chevalley–Eilenberg functors, verifies that commutative algebras and differential graded Lie algebras satisfy the reconstruction hypotheses, and states the resulting equivalence. Its final section treats quasi-coherent sheaves and representations, thereby including the material of DAG-X §2.4.

Standing conventions. Unless stated otherwise, k is a field. The arguments involving differential graded Lie algebras assume $\mathrm{char}(k) = 0$. We identify k with the Eilenberg–Mac Lane E_∞-ring Hk when using spectra. The symbol $\mathscr{S}$ denotes the ∞-category of spaces. Categories of algebras, modules, and Lie algebras are ∞-categories unless a superscript dg explicitly indicates an ordinary category of chain-level objects. A mapping space is always a derived mapping space; a pullback, pushout, limit, or colimit is taken in the indicated ∞-category.

We use homological grading, as in DAG-X. A complex has differential $d : V_i \to V_{i-1}$, and

$$(V[n])_i = V_{i-n}, \qquad d_{V[n]} = (-1)^n d_V, \qquad \pi_i V = H_i(V).$$

The dual complex is $V^\vee = \mathrm{RHom}_k(V,k)$. An object is connective if its homotopy groups vanish in negative degrees. For comparison with the cohomological convention, write $V^j = V_{-j}$; then $H^j(V) = H_{-j}(V)$. The superscript in $C^*(\mathfrak{g})$ records a cochain construction, not a change in the ambient homological convention. In particular, a classical Lie algebra concentrated in degree zero has Chevalley–Eilenberg cochains in nonpositive homological degrees.

The equivalence with differential graded Lie algebras involves all such Lie algebras after inverting quasi-isomorphisms. There is no boundedness, finite-dimensionality, or nilpotence hypothesis on the Lie algebra in the main theorem. Finiteness conditions occur on the test algebras and on the intermediate subcategory used to prove biduality. Distinguishing these different uses of finiteness is essential.

Chapter 1

Formal Moduli Problems and Abstract Deformation Theory

1.1 Infinitesimal test objects

The phrase "near a point" needs an algebraic meaning before a formal neighborhood can be studied. For ordinary affine geometry, one probes the point with local Artinian rings whose residue field is k. In derived geometry, a test algebra can also have positive homotopy groups. Lurie's definition organizes these test objects by the elementary extensions needed to build them from the residue field.

Definition 1.1.1 (Augmented algebras and square-zero objects)**.** An augmented commutative k-algebra is an object of

$$\mathrm{CAlg}_k^{\mathrm{aug}} = (\mathrm{CAlg}_k)_{/k}.$$

For a k-module M, the split square-zero algebra $k \oplus M$ has augmentation to k, and the product of two elements in its summand M is zero. The unit and augmentation make k a zero object of $\mathrm{CAlg}_k^{\mathrm{aug}}$.

Definition 1.1.2 (Deformation context; DAG-X, Definition 1.1.3)**.** A deformation context consists of a presentable ∞-category $\mathscr{C}$ and a set of spectrum objects $\{E_\alpha\}_{\alpha \in I}$ in $\mathrm{Sp}(\mathscr{C})$. Write $*$ for the terminal object of $\mathscr{C}$ and $\Omega^{\infty - n} E_\alpha$ for the nth space of E_α, regarded as an object of $\mathscr{C}$ with its specified point.

Example 1.1.3 (The commutative deformation context)**.** Take $\mathscr{C} = \mathrm{CAlg}_k^{\mathrm{aug}}$. There is a canonical

equivalence

$$\mathrm{Sp}(\mathrm{CAlg}_k^{\mathrm{aug}}) \simeq \mathrm{Mod}_k.$$

Let E correspond to k. The test objects supplied by E are

$$\Omega^{\infty-n}E \simeq k \oplus k[n].$$

Here n can be any integer as a statement about the spectrum object. Only $n \geq 0$ is used for the small test objects. See [Lur11, Example 1.1.4]

Definition 1.1.4 (Elementary and small morphisms)**.** In a deformation context, a morphism $A' \to A$ is elementary if it appears in a pullback square

$$\begin{array}{ccc} A' & \longrightarrow & * \\ \downarrow & & \downarrow \\ A & \longrightarrow & \Omega^{\infty-n}E_\alpha \end{array} \qquad (n > 0).$$

The right vertical morphism is the specified base point. A small morphism is a finite composite of elementary morphisms, including the empty composite. An object A is small if $A \to *$ is small. Let $\mathscr{C}^{\mathrm{sm}}$ be the full subcategory of small objects. These are [Lur11, Definitions 1.1.5 and 1.1.8].

Intuition. An elementary extension adds one infinitesimal direction. The map $A \to \Omega^{\infty-n}E_\alpha$ specifies the obstruction to separating that direction from A. The pullback makes the obstruction nullhomotopic. In the commutative context the added fiber is $k[n-1]$, so the positive index in the definition produces a connective fiber.

Proposition 1.1.5 (Elementary extensions of connective algebras)**.** *Let $A' \to A$ be a map of connective augmented commutative k-algebras. It is elementary if and only if its fiber, as an A'-module, is equivalent to $k[r]$ for some $r \geq 0$, with A' acting through the augmentation, and, when $r = 0$, the multiplication on $\pi_0 \operatorname{fib}(A' \to A)$ is zero.*

Proof.

Lemma 1.1.6 (Closure and size of the test category)**.** *Small morphisms are stable under composition and base change. If $A \to B$ is small and B is small, then A is small. The category $\mathscr{C}^{\mathrm{sm}}$ is essentially small, and $\Omega^{\infty-n}E_\alpha$ is small for every $n \geq 0$.*

Proof.

Proposition 1.1.7 (Recognition of small algebras)**.** *An augmented commutative k-algebra A is small if and only if all of the following hold:*

1. *$\pi_i A = 0$ for $i < 0$ and for i sufficiently large;*
2. *every $\pi_i A$ is finite-dimensional over k;*
3. *$\pi_0 A$ is a local ring whose residue field is identified with k by the augmentation.*

In particular, $\pi_0 A$ is Artinian. We write $\mathrm{Art}_k = (\mathrm{CAlg}_k^{\mathrm{aug}})^{\mathrm{sm}}$.

Proof.

Intuition. There are two finite filtrations in this criterion. Nilpotence of the maximal ideal controls the ordinary local ring. The finite Postnikov tower controls the higher homotopy groups. A finite-dimensional module over an Artinian local ring with residue field k has a finite composition series with factors k. Thus the elementary directions are exactly the pieces needed by the derived Artinian test category. A Postnikov layer need not have trivial maximal-ideal action before it is refined into those pieces.

Remark 1.1.8 (The module action in a Postnikov layer). For $B = \tau_{\leq r-1}A$ and $M = \pi_r A$, the square-zero presentation has the form

$$\begin{array}{ccc} \tau_{\leq r}A & \longrightarrow & B \\ \downarrow & & \downarrow 0 \\ B & \xrightarrow[d_r]{} & B \oplus M[r+1]. \end{array}$$

The module M is a $\pi_0 B$-module. Replacing this diagram immediately by a square over $k \oplus M[r+1]$ would lose its possibly nontrivial action. The residue-field form used in an elementary extension becomes available after isolating a composition factor with augmentation action. This distinction is useful when writing the proof of Proposition 1.1.7.

Lemma 1.1.9 (Small maps are the surjections on components)**.** *A map $A \to B$ in* Art_k *is small if and only if $\pi_0 A \to \pi_0 B$ is surjective.*

Proof.

Example 1.1.10 (Three test algebras)**.** The dual numbers $k[\varepsilon]/(\varepsilon^2)$ are $k \oplus k[0]$. The derived dual numbers $k \oplus k[r]$ with $r > 0$ have a direction in homological degree r. The ring $k[t]/(t^3)$ is a two-step infinitesimal thickening of k; its quotient to $k[t]/(t^2)$ has one-dimensional square-zero kernel. Each is an object of Art_k, although only the first two have split square-zero augmentation ideals.

1.2 The formal moduli axioms

A formal moduli problem assigns a space to each small algebra. Requiring a contractible value at the residue field chooses the object being deformed, together with an identification of its special fiber. The remaining axiom expresses the compatibility of deformations over infinitesimal gluing diagrams.

Definition 1.2.1 (Formal moduli problem; DAG-X, Definition 1.1.14)**.** For a deformation context $(\mathscr{C}, \{E_\alpha\})$, a formal moduli problem is a functor $X : \mathscr{C}^{\mathrm{sm}} \to \mathscr{S}$ such that $X(*) \simeq *$ and, for every pullback diagram of small objects

$$\begin{array}{ccc} A' & \longrightarrow & B' \\ \downarrow & & \downarrow f \\ A & \longrightarrow & B \end{array}$$

with f small, the induced map

$$X(A') \longrightarrow X(A) \times_{X(B)} X(B')$$

is an equivalence. Write $\mathrm{Moduli}_{\mathscr{C}}$ for the full subcategory of $\mathrm{Fun}(\mathscr{C}^{\mathrm{sm}}, \mathscr{S})$ consisting of these functors, and write Moduli_k in the commutative context.

Proposition 1.2.2 (Elementary tests suffice)**.** *The pullback condition in the definition of a formal moduli problem can equivalently be tested only when f is elementary, or only when f is a base point map $* \to \Omega^{\infty-n} E_\alpha$ for $n > 0$.*

Proof.

Proposition 1.2.3 (The derived Schlessinger condition)**.** *A functor $X : \mathrm{Art}_k \to \mathscr{S}$ is a formal moduli problem if and only if $X(k)$ is contractible and X preserves pullback squares in Art_k whenever at least one of the two maps to the lower-right corner is surjective on π_0. Equivalently, it suffices to require this when both maps are surjective on π_0.*

Proof.

Intuition. The axiom says that a deformation over the glued algebra is a pair of deformations together with an identification of their restrictions. The identification is part of a homotopy pullback. Passing prematurely to sets of isomorphism classes would discard the automorphisms of

this identification, and can destroy the gluing property. Nor does the axiom require preservation of every pullback: the smallness hypothesis is what makes the gluing infinitesimal.

Proposition 1.2.4 (Corepresentable problems and completion)**.** *For every object $R \in \mathscr{C}$, the functor*

$$\mathrm{Spec}(R)(A) = \mathrm{Map}_{\mathscr{C}}(R,A), \qquad A \in \mathscr{C}^{\mathrm{sm}},$$

is a formal moduli problem. More generally, if $Y : \mathscr{C}^{\mathrm{sm}} \to \mathscr{S}$ satisfies the pullback axiom and $y \in Y(*)$, then

$$Y_y(A) = Y(A) \times_{Y(*)} \{y\}$$

is a formal moduli problem.

Proof.

Proposition 1.2.5 (Presentability and formalization)**.** *The inclusion*

$$\mathrm{Moduli}_{\mathscr{C}} \hookrightarrow \mathrm{Fun}(\mathscr{C}^{\mathrm{sm}}, \mathscr{S})$$

is an accessible reflective localization. In particular, $\mathrm{Moduli}_{\mathscr{C}}$ *is presentable. Limits and filtered colimits of formal moduli problems are computed pointwise. General colimits are obtained by forming the functor-category colimit and then applying the reflector.*

Proof.

The last distinction is important in the reconstruction argument. It is easy to evaluate a limit or a filtered colimit at an algebra. A geometric realization in the category of formal moduli problems need not be the pointwise geometric realization without an additional argument. Smooth hypercoverings supply exactly such an argument in Section 1.6.

1.3 Tangent complexes and obstruction spaces

Evaluating a formal moduli problem on the dual numbers produces a pointed space. The same axiom that controls infinitesimal gluing provides compatible deloopings of that space. The tangent object is therefore naturally a spectrum; in the commutative setting it carries a k-linear structure.

Proposition 1.3.1 (The spectrum of infinitesimal directions)**.** *Let $X \in \mathrm{Moduli}_{\mathscr{C}}$. For each α and $n \geq 0$, set*

$$T_{X,\alpha,n} = X(\Omega^{\infty-n} E_\alpha).$$

There are natural equivalences $T_{X,\alpha,n} \simeq \Omega T_{X,\alpha,n+1}$, and these spaces define a spectrum $T_{X,\alpha}$. Equivalently, if E_α is regarded as a reduced excisive functor on finite pointed spaces, the composite $X \circ E_\alpha$ is reduced and excisive.

Proof.

Definition 1.3.2 (Tangent complex)**.** The spectrum $T_{X,\alpha}$ is the tangent spectrum of X in the direction E_α. In the commutative context write T_X. Its zeroth space is the tangent space $X(k \oplus k)$, and

$$X(k \oplus k[n]) \simeq \Omega^\infty(T_X[n]) \qquad (n \geq 0).$$

The designation "complex" refers to the k-module structure on T_X, not merely to an arbitrary spectrum.

Proposition 1.3.3 (Linear behavior on split extensions)**.** *For $X \in \mathrm{Moduli}_k$, the spectrum T_X has a natural k-module structure. If M is a connective k-module with finite-dimensional total homology, then*

$$X(k \oplus M) \simeq \Omega^\infty(T_X \otimes_k M).$$

Consequently, for $i, n \geq 0$,

$$\pi_i X(k \oplus k[n]) \cong \pi_{i-n} T_X.$$

Proof.

Intuition. The ordinary tangent space is only the first visible level of a single linear object. A direction in degree n shifts which homotopy group of this object is visible as a connected component. This makes it possible for an obstruction group, which is not a space of ordinary first-order deformations, to become a space of first-order deformations over a derived test algebra.

Proposition 1.3.4 (Tangent complex of a corepresentable problem)**.** *Let R be an augmented commutative k-algebra, and let $X = \mathrm{Spec}(R)|_{\mathrm{Art}_k}$. There is a natural equivalence*

$$T_X \simeq \mathrm{RHom}_R(L_{R/k}, k) \simeq (k \otimes_R L_{R/k})^\vee.$$

If R is an ordinary augmented k-algebra with augmentation ideal $\mathfrak{m}$, then $\pi_0 T_X \cong (\mathfrak{m}/\mathfrak{m}^2)^\vee$ and $\pi_i T_X = 0$ for $i > 0$.

Proof.

Proposition 1.3.5 (The obstruction fiber sequence)**.** *Suppose $A' \to A$ is an elementary map in* Art_k *with defining square*

$$A' \simeq A \times_{k \oplus k[n+1]} k, \qquad n \geq 0.$$

For every formal moduli problem X there is a fiber sequence of pointed spaces

$$X(A') \longrightarrow X(A) \longrightarrow X(k \oplus k[n+1]).$$

A point $x \in X(A)$ lifts to $X(A')$ precisely when its image lies in the base point component. When a lift and hence a nullhomotopy is chosen, the space of lifts over x is equivalent to $\Omega X(k \oplus k[n+1])$.

Proof.

The obstruction in this statement lies in $\pi_{-n-1} T_X$. In the cohomological convention for a controlling Lie algebra, the case $n = 0$ becomes an obstruction in $H^2(\mathfrak{g})$. The appearance of this group is dictated by the square-zero square and the tangent shift, rather than by an arbitrary choice of obstruction theory.

Proposition 1.3.6 (Tangent detection of equivalences)**.** *Let $u : X \to Y$ be a morphism of formal moduli problems in a deformation context. If $T_{X,\alpha} \to T_{Y,\alpha}$ is an equivalence for every α, then u is an equivalence.*

Proof.

Remark 1.3.7 (Conservative does not mean fully faithful). Proposition 1.3.6 concerns an already specified map $X \to Y$. It does not say that two formal moduli problems with isomorphic underlying tangent complexes are equivalent. The extra structure governing how tangent directions interact is precisely what the Lie bracket and its derived coherence retain. The forgetful functor from Lie algebras to complexes has the same distinction: it detects equivalences of maps, but does not reconstruct every Lie algebra from its underlying complex.

1.4 Deformation theories and reconstruction

The preceding sections describe what a formal moduli problem must do. A deformation theory is additional structure that provides algebraic objects encoding these functors. Its axioms separate the finite duality calculation from the categorical argument that extends it to all formal moduli problems.

Definition 1.4.1 (Weak deformation theory; DAG-X, Definition 1.3.1)**.** For a deformation context $(\mathscr{C}, \{E_\alpha\})$, a weak deformation theory is a functor $D : \mathscr{C}^{\mathrm{op}} \to \mathscr{D}$ with the following properties.

(D1) $\mathscr{D}$ is presentable.

(D2) D has a left adjoint $D' : \mathscr{D} \to \mathscr{C}^{\mathrm{op}}$.

(D3) There is a full subcategory $\mathscr{D}_0 \subseteq \mathscr{D}$ such that:

- (a) $K \to DD'K$ is an equivalence for every $K \in \mathscr{D}_0$;
- (b) the initial object 0 belongs to $\mathscr{D}_0$;
- (c) for each $n > 0$ and α there is $K_{\alpha,n} \in \mathscr{D}_0$ with $D'K_{\alpha,n} \simeq \Omega^{\infty-n}E_\alpha$;
- (d) $\mathscr{D}_0$ is closed under pushouts $K \sqcup_{K_{\alpha,n}} 0$, where $K \in \mathscr{D}_0$ and the map $K_{\alpha,n} \to 0$ is induced by the specified base point.

The adjunction has a variance that will be used repeatedly:

$$\mathrm{Map}_{\mathscr{D}}(K, D(A)) \simeq \mathrm{Map}_{\mathscr{C}}(A, D'K).$$

The counit in $\mathscr{C}^{\mathrm{op}}$ becomes a map $A \to D'D(A)$ in $\mathscr{C}$. By contrast, the formal moduli problem associated to K will use maps $D(A) \to K$. These two mapping-space formulas must not be interchanged.

Intuition. The subcategory $\mathscr{D}_0$ is a region where duality is controlled. The elementary spheres must have duals there, and attaching the dual of an elementary extension must stay there. This

is enough to recover every small test algebra. The theorem does not ask for duality to be an equivalence between the entire categories $\mathscr{C}^{\mathrm{op}}$ and $\mathscr{D}$.

Proposition 1.4.2 (Consequences of weak duality)**.** *For a weak deformation theory, $D(*)$ is initial. If A is small, then $D(A) \in \mathscr{D}_0$ and the canonical map*

$$A \longrightarrow D'D(A)$$

is an equivalence. Moreover, D takes a pullback square with small lower objects and a small right vertical map to a pushout square in $\mathscr{D}$.

Proof.

Corollary 1.4.3 (The comparison functor)**.** *Every weak deformation theory determines a functor*

$$\Psi : \mathscr{D} \longrightarrow \mathrm{Moduli}_{\mathscr{C}}, \qquad \Psi(K)(A) = \mathrm{Map}_{\mathscr{D}}(D(A), K).$$

Proof.

Definition 1.4.4 (Deformation theory; DAG-X, Definition 1.3.9)**.** For a weak deformation theory, put $e_\alpha(K) = T_{\Psi(K),\alpha}$. It is a deformation theory if it also satisfies:

(D4) Each $e_\alpha : \mathscr{D} \to \mathrm{Sp}$ preserves small sifted colimits, and the family $\{e_\alpha\}$ is jointly conservative.

Sifted colimits include filtered colimits and geometric realizations of simplicial objects. Preserving filtered colimits is enough to control passage from finite tests to prorepresentable functors. Preserving geometric realizations is needed to assemble a smooth resolution. The second requirement in (D4) says that the resulting algebraic objects have no invisible component beyond what their tangent spectra can detect.

Theorem 1.4.5 (Abstract reconstruction; DAG-X, Theorem 1.3.12)**.** *If $D : \mathscr{C}^{\mathrm{op}} \to \mathscr{D}$ is a deformation theory, then*

$$\Psi : \mathscr{D} \xrightarrow{\simeq} \mathrm{Moduli}_{\mathscr{C}}, \qquad K \longmapsto \big(A \longmapsto \mathrm{Map}_{\mathscr{D}}(D(A), K)\big)$$

is an equivalence of ∞-categories.

Proof.

Intuition. Small algebras form the finite pieces of the theory. Prorepresentable objects allow coherent limits of such pieces, and smooth resolutions allow higher gluing. The axioms of a deformation theory ensure that the algebraic category handles both operations and that tangent spectra recognize the resulting comparison. The theorem is a reconstruction statement about full spaces of morphisms, not merely a bijection between isomorphism classes.

Corollary 1.4.6 (Corepresentables under the equivalence)**.** *For every $A \in \mathscr{C}$, there is a natural equivalence $\Psi(D(A)) \simeq \mathrm{Spec}(A)$ on $\mathscr{C}^{\mathrm{sm}}$. In particular, a deformation theory, when it exists, identifies the functor D with the restriction-of-corepresentables construction under the equivalence of Theorem 1.4.5.*

Proof.

1.5 The small object argument

The proof of abstract reconstruction needs resolutions with prescribed lifting properties. Lurie's small object argument is carried out directly in a presentable ∞-category. Its lifting condition asks for the existence of coherent fillers; it does not require the space of fillers to be contractible.

Definition 1.5.1 (Lifting and cellular maps)**.** For morphisms $f : C \to D$ and $g : X \to Y$ in an

∞-category, say that g has the right lifting property against f if the map

$$\mathrm{Map}(D,X) \longrightarrow \mathrm{Map}(C,X) \times_{\mathrm{Map}(C,Y)} \mathrm{Map}(D,Y)$$

is surjective on connected components. A transfinite cellular composite generated by a set S is an ordinal-indexed composite of pushouts of maps in S, with colimits taken at limit ordinals. A class of maps is weakly saturated if it is closed under these composites and retracts.

Proposition 1.5.2 (Small object factorization)**.** *Let $\mathscr{E}$ be presentable and let S be a small set of morphisms in $\mathscr{E}$. Every map $f : X \to Z$ has a factorization $X \to Y \to Z$ whose first map is a transfinite cellular composite generated by S and whose second map has the right lifting property against S.*

Proof.

Lemma 1.5.3 (Weak saturation of a lifting class)**.** *The maps having the left lifting property against a fixed collection of morphisms form a weakly saturated class. If S is a small set in a presentable category, its weakly saturated closure is the class of maps having the left lifting property against every map with the right lifting property against S.*

Proof.

Definition 1.5.4 (Relative latching and matching objects)**.** For a simplicial object $U_\bullet$ of $\mathscr{E}_{/X}$, the latching object L_nU is the colimit of its degeneracy data into degree n. The matching object M_nU is the limit of the already specified faces out of degree n, computed in $\mathscr{E}_{/X}$. There is a canonical map $L_nU \to M_nU$. In particular, M_0U is the terminal object of $\mathscr{E}_{/X}$, namely $X \xrightarrow{\mathrm{id}} X$.

Proposition 1.5.5 (Simplicial resolution with lifting maps)**.** *For presentable $\mathscr{E}$, a small set S of maps, and $X \in \mathscr{E}$, there is a simplicial object $U_\bullet$ in $\mathscr{E}_{/X}$ such that each underlying map $L_nU \to U_n$ is cellular with respect to S, and each underlying map $U_n \to M_nU$ has the right lifting property against S.*

Proof.

Intuition. The latching object records identifications already forced by degeneracies. The matching object records compatible boundary data that a new simplex must fill. Factoring the map between them supplies a new level without discarding the choices made in lower degrees. In the application to moduli problems, the matching maps become smooth, and the cellular maps keep the levels prorepresentable.

1.6 Prorepresentability and smooth hypercoverings

The passage from finite algebras to complete local behavior is expressed by pro-objects. It is helpful to distinguish the inverse system of algebras from the direct system of functors it represents.

Definition 1.6.1 (Prorepresentable functors)**.** A functor $X : \mathscr{C}^{\mathrm{sm}} \to \mathscr{S}$ is prorepresentable if it is a filtered colimit of corepresentables $\mathrm{Spec}(A_i)$ with $A_i \in \mathscr{C}^{\mathrm{sm}}$. Equivalently, it comes from a pro-object of $\mathscr{C}^{\mathrm{sm}}$, viewed contravariantly as a functor of points. For a cofiltered inverse system $\{A_i\}_{i\in I}$ the formula is

$$X(B) \simeq \operatorname*{colim}_{i\in I^{\mathrm{op}}} \mathrm{Map}_{\mathscr{C}}(A_i, B).$$

Proposition 1.6.2 (Prorepresentability is compatible with small cells)**.** *Every prorepresentable functor is a formal moduli problem. If a map $X \to Y$ in* $\mathrm{Moduli}_{\mathscr{C}}$ *is a transfinite cellular composite of maps* $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ *induced by small maps $A \to B$ of small objects, and X is prorepresentable, then Y is prorepresentable.*

Proof.

Bibliography

[Get09] Ezra Getzler, *Lie theory for nilpotent L_∞-algebras*, Annals of Mathematics (2) **170** (2009), no. 1, 271–301.
https://doi.org/10.4007/annals.2009.170.271.

[Lur11] Jacob Lurie, *Derived Algebraic Geometry X: Formal Moduli Problems*, November 5, 2011.
https://www.math.ias.edu/~lurie/papers/DAG-X.pdf.

[Lur17a] Jacob Lurie, *Higher Algebra*, version of September 18, 2017.
https://www.math.ias.edu/~lurie/papers/HA.pdf.

[Lur17b] Jacob Lurie, *Higher Topos Theory*, Annals of Mathematics Studies, vol. 170, Princeton University Press, 2009; online version of April 9, 2017.
https://www.math.ias.edu/~lurie/papers/HTT.pdf.